Octane Obit

Last mile fuel management solution

Solution description Plan

Lilian Musyoka : Founder Octane Orbit
Tel *: +254 701 321 061*
Email *: summitsurdsinternational@gmail.com*
Email *: lilian@summitsurds.com*

Solving the retail tail end of fuel ecosystem from the depot, to sales , to bank

01 EXECUTIVE SUMMARY

Overview

Octane Obit is an **end-to-end downstream fuel operations platform** designed to manage the **last mile of fuel and product movement** — from **depot delivery**, through **station dispensing**, to **sales reconciliation and banking**.

The system is built to eliminate the traditional blind spots in fuel operations by tightly coupling **physical flow** (litres moving through meters and nozzles) with **financial flow** (sales values, expected revenue, and banked amounts).

Octane Obit does not just record transactions — it continuously answers three critical operational questions:

Lilian Musyoka : Founder Octane Orbit
Tel *: +254 701 321 061*
Email *: summitsurdsinternational@gmail.com*
Email : lilian@summitsurds.com

1. **What should have happened?** (Expected delivery, expected sales, expected cash)
2. **What actually happened?** (Delivered fuel, dispensed fuel, recorded sales, banked amounts)
3. **What is the variance — and where did it occur?**

This document explains each major capability and module that supports this full lifecycle, as represented in the sidebar navigation configuration.

02 PROBLEM OVERVIEW

Fuel operations suffer from a critical visibility gap between **physical fuel movement** and **financial outcomes**. Most stations rely on paperwork, manual reconciliations, or disconnected systems to track deliveries, sales, and banking. This creates an environment where losses are detected late — or not at all.

Common pain points include:

- Fuel deliveries confirmed by paperwork rather than measured reality
- Inability to prove short deliveries or receiving losses

- Pump and nozzle sales that cannot be reliably reconciled against tank stock
- Cash sales that do not fully translate into banked amounts
- End-of-day reconciliation that depends on trust instead of data

In high-volume stations, even small percentage losses compound quickly. A **1–3% leakage** across delivery, dispensing, or banking can translate into **millions in annual revenue loss**, while remaining invisible within manual or fragmented systems.

03

SOLUTION OVERVIEW

Octane Obit reintroduces **control, measurement, and accountability** into fuel and retail operations by unifying delivery, dispensing, sales, and banking into a single operational truth.

The platform uniquely solves these challenges by:

- Measuring fuel **as it flows** during delivery using flow meters

- Tying every wet sale to a **physical nozzle and pump**
- Continuously computing **expected vs actual** values across stock, sales, and cash
- Automatically surfacing **variances at the exact point they occur**

Rather than relying on periodic audits, Octane Obit enables **continuous reconciliation** — allowing operators and management to act in real time, not after losses have already accumulated.

End-to-End Operational Flow (High Level)

Octane Obit operates on a closed-loop accountability model:

Depot → Delivery → Tank → Pump/Nozzle → Sale → Cash/Bank

At every transition point, the system captures:

- **Expected quantity/value**
- **Actual measured quantity/value**
- **Variance (loss, gain, delay, or mismatch)**

This creates a continuous audit trail that connects physical fuel movement to financial outcomes.

04

NOZZLE-LEVEL SALES ACCOUNTABILITY

Every wet sale in Octane Obit is tied to a **specific pump and nozzle**.

Capabilities:

- Litres dispensed are captured per nozzle
- Sales value is computed from actual dispensed volume
- Each nozzle contributes to shift-level and day-level totals

This enables:

- Attendant accountability
- Pump performance analysis
- Early detection of meter drift or leakage

05

REVENUE MONITORING CAPABILITY

Purpose: Ensures real-time financial accountability by reconciling sales against actual banked and deposited amounts at the end of every shift.

This module is designed to eliminate delayed discovery of cash discrepancies by enforcing **shift-level reconciliation before operational handover**.

1 Digital Payment Collection Tracking

Octane Obit automatically tracks all **digital payment modes** processed through the system, including:

- Mobile money
- Card payments
- Bank transfers
- Any configured electronic payment channels

For each shift, the system already knows:

- Total sales value
- Breakdown by payment method
- Expected non-cash collections

This creates a clear, system-verified **expected digital collection total** without manual computation.

---

2 Cash Deposit Declaration & Validation

At the end of shift, the responsible supervisor or manager is required to **declare the actual cash deposited or prepared for banking**.

Octane Obit then:

- Compares declared cash deposits against **expected cash sales**
- Automatically computes surplus or shortage
- Records variance with timestamps and responsible shift metadata

This removes ambiguity and prevents retrospective adjustments.

---

3 Shift Closure Enforcement (Control Gate)

A core control feature of Octane Obit is **conditional shift progression**.

- If sales and banking figures reconcile within acceptable thresholds, the shift is closed successfully.
- If discrepancies exist, the system flags the variance and **blocks access to the next operational shift** until reconciliation is resolved or approved.

This enforces accountability **in real time**, rather than discovering discrepancies days or weeks later.

---

4 Real-Time Variance Visibility

All banking variances are:

- Immediately visible to management
- Linked to specific shifts, users, and dates
- Available for audit and escalation workflows

By surfacing discrepancies at the moment they occur, Octane Obit transforms banking from a delayed accounting exercise into an **operational control mechanism**.

06

PRODUCT DELIVERIES AND SALES MONITORING

- Expected vs Actual Sales Logic

Once delivery establishes stock levels, Octane Obit continuously computes:

- **Expected sales** (based on delivered fuel and opening balances)
- **Actual sales** (from nozzle and POS data)

The system highlights variances such as:

- Fuel dispensed but not sold
- Sales recorded without corresponding dispensing
- Timing mismatches between shifts

This transforms sales data from "records" into **control signals**.

Lilian Musyoka : Founder Octane Orbit
Tel *: +254 701 321 061*
Email *: summitsurdsinternational@gmail.com*
Email *: lilian@summitsurds.com*

07

MERCHANT ONBOARING WORKFLOWS

Octane Obit is designed with **rapid merchant onboarding** as a core capability, recognizing that many stations and fuel businesses already operate with historical data from legacy systems, third-party software, or spreadsheet-based workflows. The platform includes structured data setup and migration workflows that allow merchants to be onboarded quickly without disrupting ongoing operations.

The system supports onboarding from **existing digital sources** such as Excel files, CSV exports, or database dumps from older systems. Core datasets — including products, fuel types, opening stock balances, clients, vehicles, pricing structures, and outstanding balances — can be validated, normalized, and mapped into Octane Obit's data model. This ensures continuity of operations while preserving historical accuracy.

During onboarding, Octane Obit establishes a clear **baseline state**: expected stock, expected balances, and operational starting points.

This approach significantly reduces time-to-value for new merchants, enabling them to transition from legacy tools to a **fully controlled, variance-aware**

operational environment in days rather than months, while maintaining trust in their historical and financial data.

SYSTEM MODULES OVERVIEW

1. Telematics & Asset Visibility

Telematics in Octane Obit goes beyond just product tracking — it forms a **fuel integrity monitoring layer** that continuously validates fuel movement and storage using physical measurement devices.

This layer combines **underground tank monitoring**, **delivery flow measurement**, and **nozzle-level dispensing data** to detect losses, anomalies, and acceptable operational shrinkage in real time.

Key Components & Capabilities:

- **Underground Tank Monitoring (Digital Submersibles / Probes):**

    - Continuous measurement of tank levels, temperature, and volume
    - Detection of unexplained drops, slow leaks, or abnormal tank behavior
    - Correlation of tank readings with restock and dispensing events
- **Delivery Flow Meter Integration:**
    - Real-time measurement of fuel as it is offloaded during depot deliveries
    - Independent verification of delivery notes and supplier declarations
    - Timestamped linkage between delivery start/end and tank level increases
- **Nozzle & Pump Flow Meter Monitoring:**
    - Measurement of actual fuel dispensed during sales
    - Correlation between nozzle totals and tank depletion
    - Early detection of meter drift, calibration issues, or losses between tank and pump
- **Fuel Chemistry & Acceptable Loss Modeling:**
    - Accounts for industry-accepted fuel behavior such as temperature expansion, contraction, evaporation, and handling losses
    - Defines **acceptable loss thresholds** based on fuel type and operating conditions
    - Automatically distinguishes between normal technical losses and suspicious variance

By correlating **tank levels → delivered volume → dispensed volume → sold value**, Octane Obit can accurately determine:

- Where fuel was lost
- Whether the loss is within acceptable chemical and operational limits
- When intervention or investigation is required

This transforms telematics from simple tracking into a **fuel loss detection and prevention system**, making Octane Obit a true **end-to-end fuel integrity and control platform**.

2. Flow Meter–Driven Delivery Verification

During fuel delivery from depot to station, Octane Obit integrates a **flow meter–based delivery capture process**.

What this means:

- Delivered fuel is measured **in real time as it flows**, not estimated from paperwork.
- The system records:
  - Expected delivery quantity (from delivery note / order)
  - Actual delivered litres (from flow meter)
  - Delivery duration and timestamps
  - Variance between expected vs delivered

This immediately exposes:

- Short deliveries
- Over-deliveries
- Meter tampering attempts
- Human error at receiving stage

This delivery data becomes the **baseline stock truth** for all downstream sales and reconciliation.

Lilian Musyoka : Founder Octane Orbit
***Tel** : +254 701 321 061*
***Email** : summitsurdsinternational@gmail.com*
***Email** : lilian@summitsurds.com*

2. Inventory Management Module

Purpose: Controls how fuel and non-fuel products are replenished, validated, and introduced into operational stock, ensuring that every increase in stock is traceable, measured, and reconciled.

Restock is a critical control point because it defines the **starting truth** from which sales, variances, and banking accuracy are derived.

Key Capabilities:

- **Fuel Restock (Wet Stock):**
  - Linked directly to depot deliveries and flow-meter–verified quantities
  - Expected delivery vs actual received fuel is captured before stock is updated

    - Prevents inflated stock balances caused by paperwork-only confirmations
- **Product Restock (Dry Stock):**
    - Supports manual and bulk restocking of shop items and accessories
    - Quantity, cost price, and supplier references are captured
    - Ensures cost-accurate margin computation downstream
- **Baseline Enforcement:**
    - Restock operations establish clean opening balances
    - All downstream sales and reconciliation depend on this baseline

This module ensures that **no stock enters the system without measurement, validation, and accountability**.

3. Sales Module

Purpose: Handles all revenue-generating transactions, both fuel and non-fuel, from point-of-sale to reporting.

1.1 Add Dry Sale

- **Description:** Record sales for non-fuel items such as lubricants, spare parts, shop items, or accessories.
- **Typical Users:** Cashiers, sales attendants
- **Key Features:**
    - Manual item selection

- Quantity and price entry
- Instant total calculation
- Receipt generation

1.2 Add Wet Sale

- **Description:** Record fuel sales dispensed through pumps.
- **Typical Users:** Pump attendants
- **Key Features:**
    - Pump/nozzle-based entry
    - Litres and amount tracking
    - Preset or manual fueling
    - Integration with fuel stock and pump logic

1.3 Sales Summary

- **Description:** High-level overview of sales performance over time.
- **Key Metrics:**
    - Total sales value
    - Fuel vs dry sales breakdown
    - Daily, weekly, and monthly aggregation

Lilian Musyoka : Founder Octane Orbit
Tel *: +254 701 321 061*
Email *: summitsurdsinternational@gmail.com*
Email *: lilian@summitsurds.com*

- **Audience:** Supervisors, managers

1.4 Sale Orders

- **Description:** Manages pending, completed, or invoiced sale orders.
- **Use Cases:**
  - Credit sales
  - Corporate or fleet clients
  - Deferred payment tracking

1.5 Dry Sales History

- **Description:** Historical log of all completed non-fuel sales.
- **Features:**
  - Date-based filtering
  - Client-based lookup
  - Audit and reconciliation support

1.6 Fuel Sales History

- **Description:** Historical log of all fuel (wet) sales transactions.

- **Features:**
    - Pump/nozzle reference
    - Litres vs amount comparison
    - Shift and attendant accountability

---

3. Clients Module

Purpose: Manages customer records and their associated assets, particularly vehicles.

2.1 Add Client

- **Description:** Create a new client profile.
- **Client Types:**
    - Walk-in customers
    - Credit clients
    - Corporate / fleet clients
- **Stored Data:**
    - Contact details
    - Credit limits
    - Billing preferences

2.2 Manage Clients

- **Description:** View, edit, and manage all registered clients.
- **Key Features:**
  - Client search and filtering
  - Credit balance tracking
  - Client activity overview

2.3 Add Client Vehicle

- **Description:** Register vehicles associated with a specific client.
- **Use Cases:**
  - Fleet fueling
  - Vehicle-based fuel limits
- **Captured Details:**
  - Registration number
  - Vehicle type
  - Assigned client

2.4 Manage Client Vehicles

- **Description:** View and manage all registered client vehicles.
- **Benefits:**
  - Fuel accountability per vehicle
  - Fraud and misuse reduction

4. Expenses Module

Purpose: Tracks operational costs to support profitability and financial control.

3.1 Add Expense

- **Description:** Record operational expenses.
- **Expense Examples:**
  - Utilities
  - Maintenance
  - Salaries
  - Supplies
- **Key Fields:**
  - Expense category
  - Amount
  - Date
  - Remarks

3.2 Manage Expenses

- **Description:** Review and manage recorded expenses.
- **Features:**
  - Period filtering
  - Category summaries

- Export and reporting support

Design Philosophy

- **Operational first:** Built for real-world station workflows
- **Modular:** Each module can evolve independently
- **Audit-ready:** Strong emphasis on traceability and history
- **Scalable:** Suitable for single stations or multi-stations / site deployments

In summary

OctaneObit is not simply a sales or station management system — it is a **real-time operational control platform** for fuel and retail businesses.

By tightly linking **delivery measurement, restocking, dispensing, sales, and banking**, the system eliminates blind spots where losses traditionally hide. Every litre and every currency unit is tracked from expectation to reality, with variances exposed immediately rather than discovered retrospectively.

The result is a platform that enforces accountability at the moment decisions are made — not after reports are reviewed. For operators, this means reduced losses and stronger control. For management and investors, it means **predictable operations, trustworthy data, and scalable governance** across single or multi-site deployments.

Lilian Musyoka : Founder Octane Orbit
Tel *: +254 701 321 061*
Email *: summitsurdsinternational@gmail.com*
Email *: lilian@summitsurds.com*

OctaneObit turns fuel operations from a trust-based process into a **measured, reconciled, and controlled system** — in real time.

Lilian Musyoka : Founder Octane Orbit
Tel *: +254 701 321 061*
Email *: summitsurdsinternational@gmail.com*
Email : lilian@summitsurds.com